25901742

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/MPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMOS100 – FOUNDATIONS OF MOBILE DEVELOPMENT SYSTEMS

(2024 ADMISSION ONWARDS)

VALUATION SCHEME

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(|) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions
Answer all questions
Each question carries 1 marks

Q.No	Answer	Level	CO
1	a) To perform a logical AND operation	K	CO1
2	c) Map	K	CO1
3	c) var map = {'key': value};	U	CO1
4	d) putString()	K	CO2
5	d) Follow the recommendations provided by flutter doctor	U	CO2
6	a) Ctrl + Shift + P	K	CO2
7	c) Firestore allows more flexible querying and scaling	U	CO3
8	d) Dart	K	CO1
9	c) Managing user sign-in and sign-up	K	CO3
10	b) To apply partial updates to a resource	U	CO3

[1x10 = 10]

Part B

Short Answer Questions
Answer any 4 questions
Each question carries 5 marks

Q11. (U)/ [CO1|]

- Meaning of exception handling → 1 mark
- Role of try block → 1 mark
- Role of catch block → 1 mark
- Role of finally block → 1 mark
- Flutter-related example explanation → 1 mark

Q12. (U)/ [CO1|]

- Definition of abstraction → 1 mark
- Use of abstract classes/interfaces → 1 mark
- How it hides complexity → 1 mark
- Importance in Flutter apps → 1 mark
- Example explanation → 1 mark

Q13. (K)/ [CO2|]

- Correct identification of the library (**Cupertino**) → 3 mark
- Recall of related fact (used for iOS-style UI / Apple design) → 1 mark
- Mention of example widget (e.g., Cupertino Button, Cupertino Navigation Bar) → 1 mark

Q14. (U)/ [CO2|]

- Meaning of release mode → 1 mark
- Meaning of debug mode → 1 mark
- Key differences explained (performance, debugging, size, etc.) → 2 marks
- Practical understanding/use case → 1 mark

Q15. (U)/ [CO3|]

- Definition of Firestore → 1 mark
- Nature (NoSQL, cloud database) → 1 mark
- Key features (real-time updates, scalability, etc.) → 2 marks
- Example usage in Flutter → 1 mark

Q16. (U)/ [CO3|]

- Definition of SQLite → 1 mark
- Role as local database → 1 mark
- Key features (offline storage, lightweight, etc.) → 2 marks
- Example use case → 1 mark

[5x4 = 20]

Part C
Essay Questions
Answer any 2 questions
Each question carries 10 marks

Q17. (U)/ [CO1|]

- Meaning of objects in Dart → **2 marks**
- Role in storing and organizing data → **2 marks**
- How objects simplify data handling in apps → **2 marks**
- Interaction with UI (widgets using object data, state handling) → **2 marks**
- Example (model class used in UI) → **2 marks**

Q18. (U)/ [CO1|]

- Explanation of for loop → **2 marks**
- Explanation of while loop → **2 marks**
- Explanation of do-while loop → **2 marks**
- Key differences (entry vs exit control, usage scenarios) → **2 marks**
- Example illustrating loops → **2 marks**

Q19. (U)/ [CO2|]

- Explanation of Material widgets → **2 marks**
- Explanation of Cupertino widgets → **2 marks**
- Differences (design style, platform usage, UI behaviour) → **3 marks**
- Appropriate use cases → **2 marks**
- Example widgets → **1 mark**

Q20. (U)/ [CO3|]

- Correct definition of **Idempotent methods → 3 marks**
- Correct definition of **Non-idempotent methods → 3 marks**
- Listing correct examples → **2 marks**
- Stating the key difference (same result vs different result on repetition) → **2 marks**

[10x2 = 20]